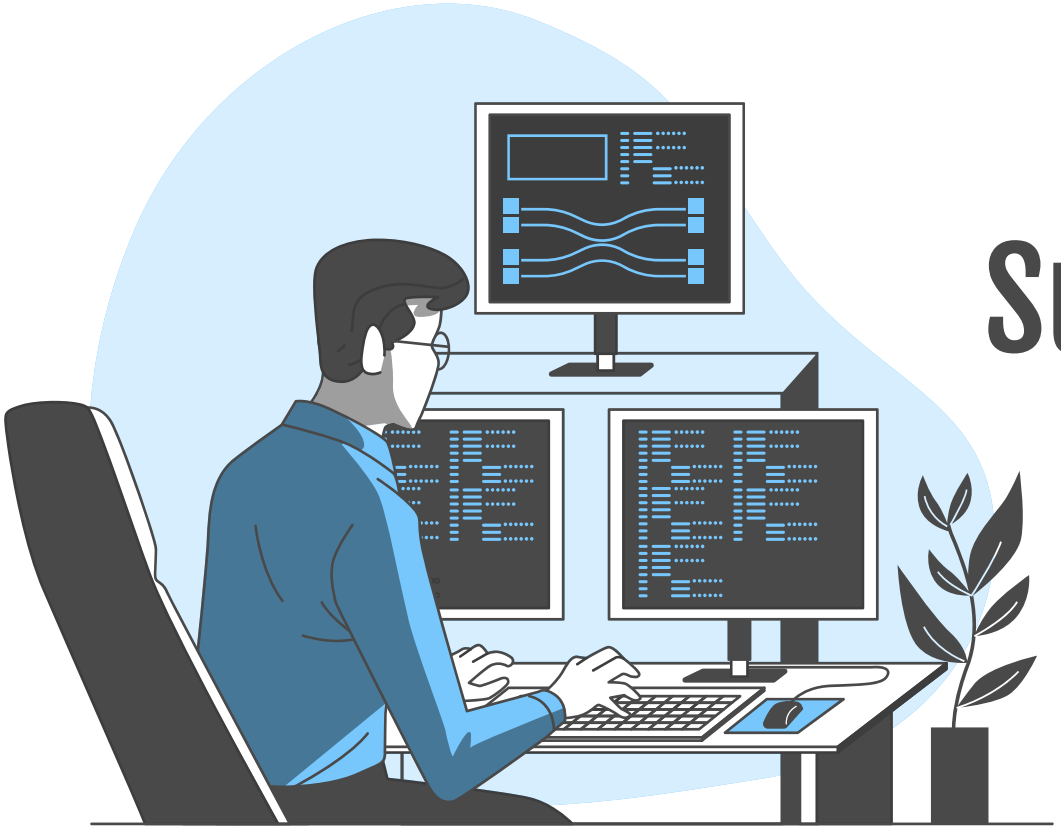


Supercomputing Robotics

A journey from laptop to datacenter and everything in between

Thijn Bakker – Third Year Internship – BUas: Data Science & Artificial Intelligence

Table of Contents

01 **Who I Was**
Confused, laptop-focused, no HPC experience

02 **Where I Went**
TU/e Supercomputing Centre

03 **What I Built**
Three projects, three audiences

04 **What Went Wrong**
And how I fixed it

05 **What I Learned**
Lessons from the datacenter

06 **Spreading the Word**
Four presentations, prevent shadow IT

07 **Impact & What's Next**
Results and legacy

01

Who I Was

Confused, laptop-focused, no HPC experience

Before the Internship: A Student with Questions

Starting point
Data Science Student

But
No HPC Experience

Knowledge Gap #1
What is the Difference?

Knowledge Gap #2
Why use a Supercomputer?

My Laptop: 8 cores, 16 GB RAM, 1 GPU vs **SPIKE-1:** 112 cores, 2 TB RAM, 8× B200 GPUs

02

Where I Went

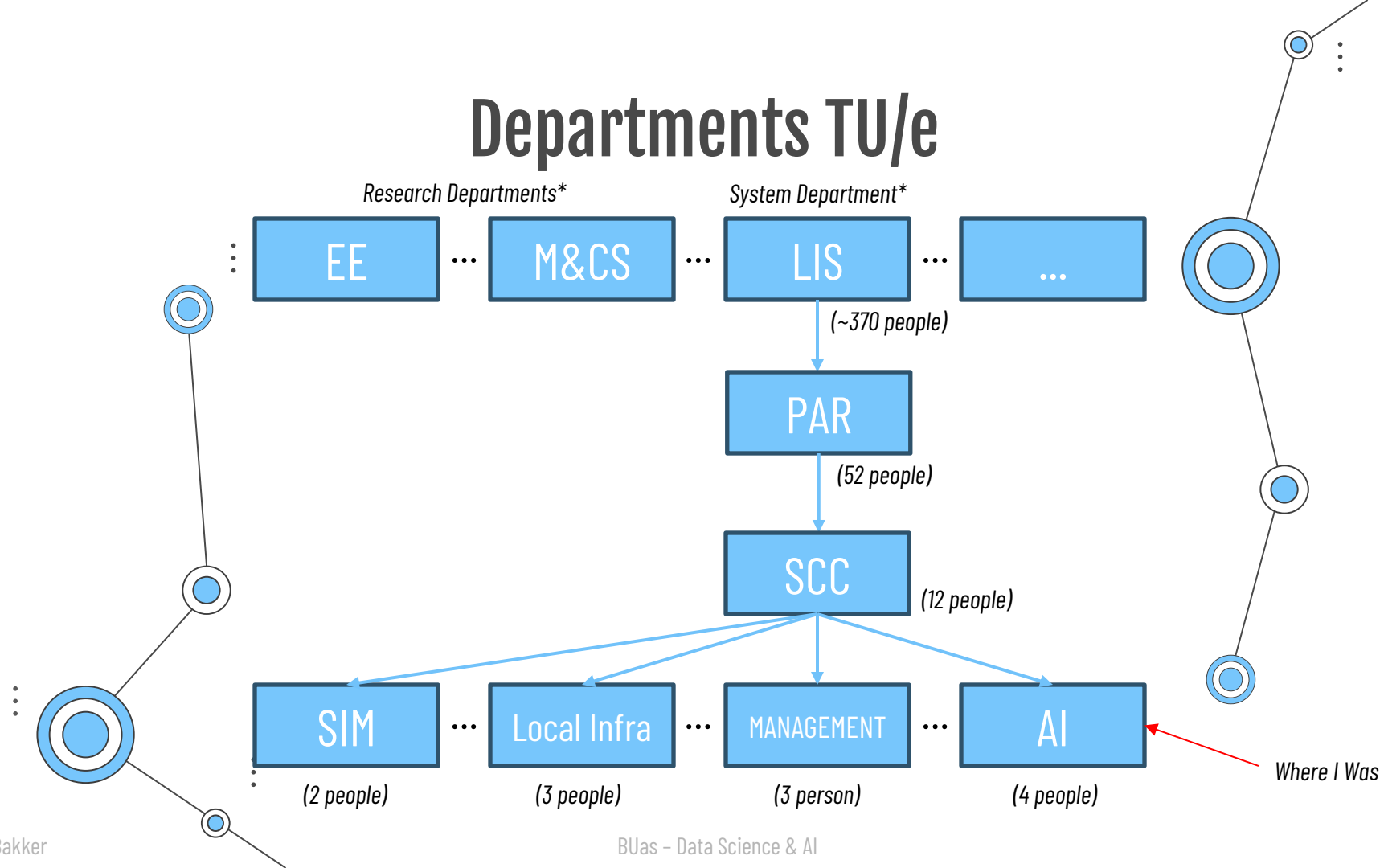
TU/e Supercomputing Centre

The Location

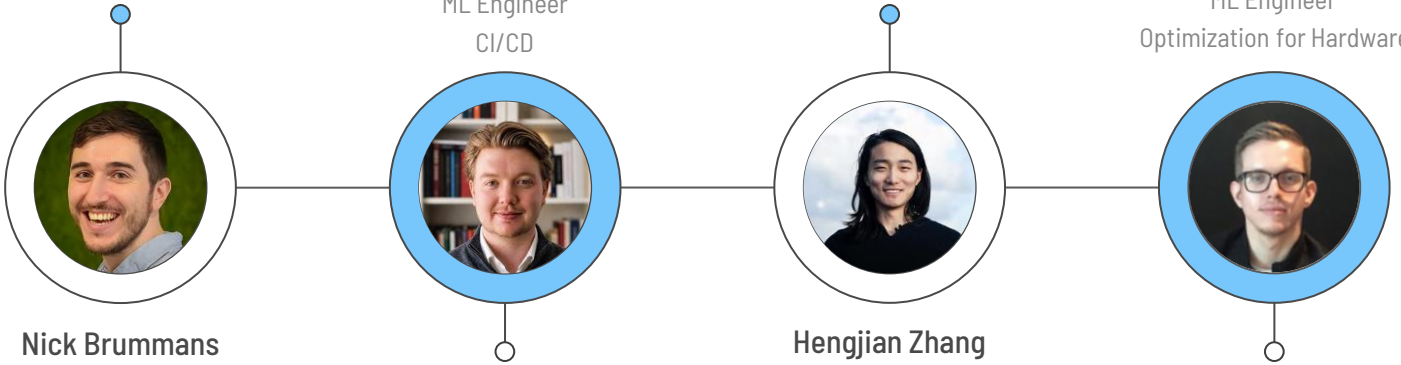




Departments TU/e



The AI Team *(4 people)*



Nick Brummans
Product Owner
Team Coordination

Bas Maat
ML Engineer
CI/CD

Hengjian Zhang
ML Engineer
Software Service Management

Bram van Berlo
ML Engineer
Optimization for Hardware

Small Team, Big Vision: "Compute to Compete"

Don't own one supercomputer, help researchers connect to MANY

International HPC — EuroHPC LUMI

National SURF — Snellius, Research Cloud

Institutional TU/e — SPIKE-1, DGX Spark

Departmental Clusters — Umbrella HPC

Laptops / Workstations — Where most researchers start

03

What I Built

Three projects, three audiences

Three Projects, Three Stories



Cross-Platform Benchmarking

"Which platform should researchers use?"

...



Isaac Sim Deployment

"How do I train robots on SPIKE-1?"

...



GAIA Team Support

"How do we (industrial startup) get started with HPC?"

...

Benchmark Results: Which Platform Wins?

Benchmark	SPIKE-1 B200 · 183 GB	Research Cloud A10 · 24 GB	DGX Spark GB10 · 125 GB	Why?
RL Training (32K envs, headless)	**385K FPS**	184K FPS	125K FPS	B200: **raw GPU compute** dominates. All 3 run 32K — fair comparison.
RL Training (16K envs, GUI)	**70K FPS**	47K FPS	55K FPS	Spark beats A10 in GUI — **unified memory** helps with render+physics.
Scene Rendering (Warehouse, headless)	**510 FPS**	31 FPS	72 FPS	**B200: 5th-gen RT cores**. A10 is compute-focused (fewer RT units).
CPU Physics (10 robots, render=False)	60 FPS	78 FPS	**122 FPS**	**Spark: unified memory** = zero PCIe copy overhead every physics step.
GPU Physics (10K rigid cubes)	376 FPS	475 FPS	**542 FPS**	Same effect: **unified memory** wins. GPU crossover at ~2K–10K objects.

Key Insight: No single platform is "best" — it depends on your workload, budget, and timeline. These results help researchers choose.

Isaac Sim Deployment: Making Kayden Independent



Researcher:
Kayden Knapik (TU/e Mechanical Engineering, Master's)
Robots:
BoosterT1 humanoid + BDX-R quadruped
Performance:
381K physics steps/sec on 3× B200 GPUs — convergence in hours, not days
Impact:
Kayden iterates rapidly, no SCC team dependency

Real Result: GPU-Accelerated RL in Action



381K steps/sec on 3× B200 GPUs | **Hours to converge** (vs. days on CPU)

GAIA Team Support: Industrial Robotics on Research Cloud

Who: TeamGaia.nl

Startup building an autonomous trash cleaning robot

What I Did:

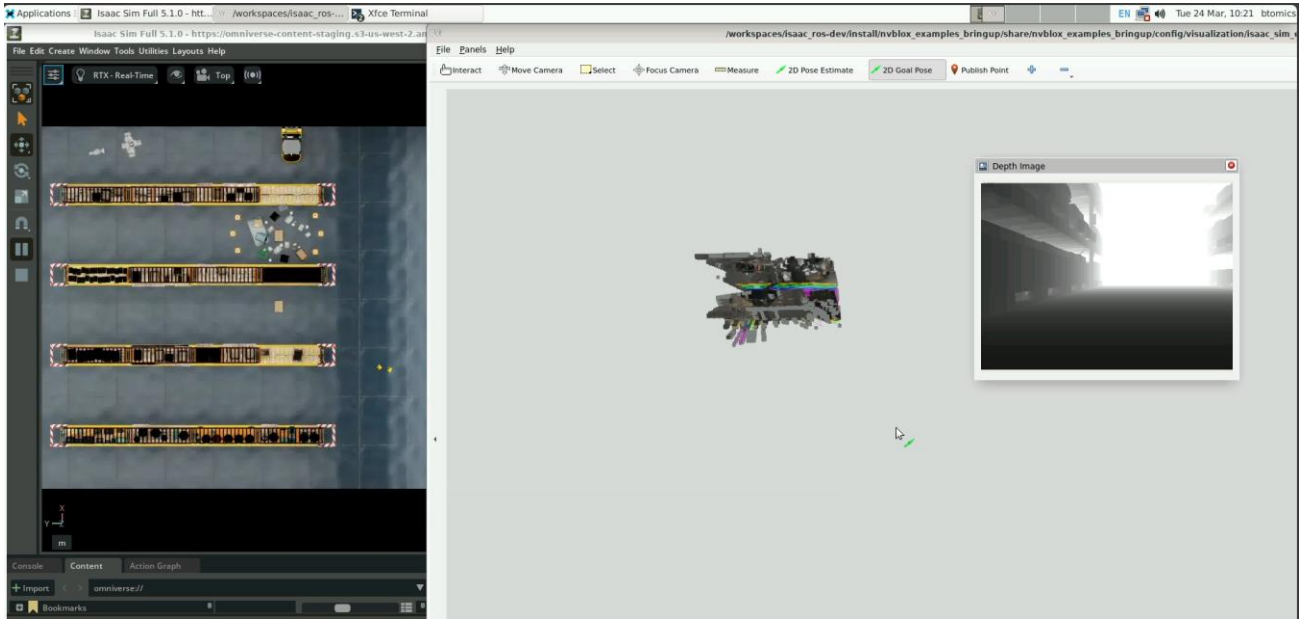
- Onboarded them on SURF Research Cloud (A10)
- Helped set up Isaac Sim for robot simulation
- Documented issues + solutions
- Integrated 3D visualization + navigation (IsaacRos)
- Provided hands-on support throughout

Their Progress:

- Collaborated with 10 people on one system
- Are seamlessly growing
- 3D visualization + navigation from IsaacRos working

Lesson: Real robotics = iteration + creative constraints.
They helped me, I helped them.

Real Result: Navigation from IsaacRos in Action



VIDIA's GPU-accelerated 3D reconstruction library (Nvblox) inside the Isaac ROS Docker container

04

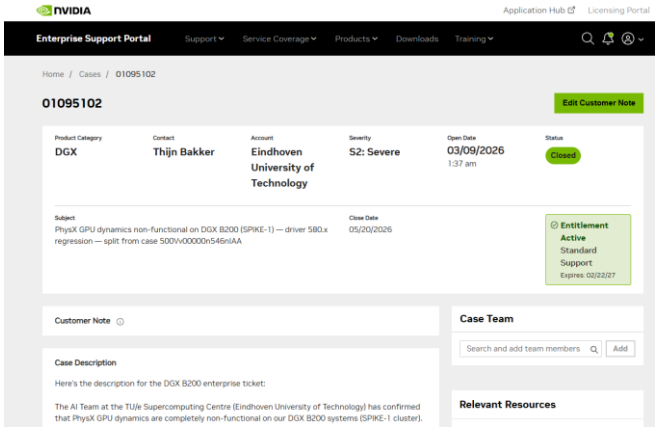
What Went Wrong

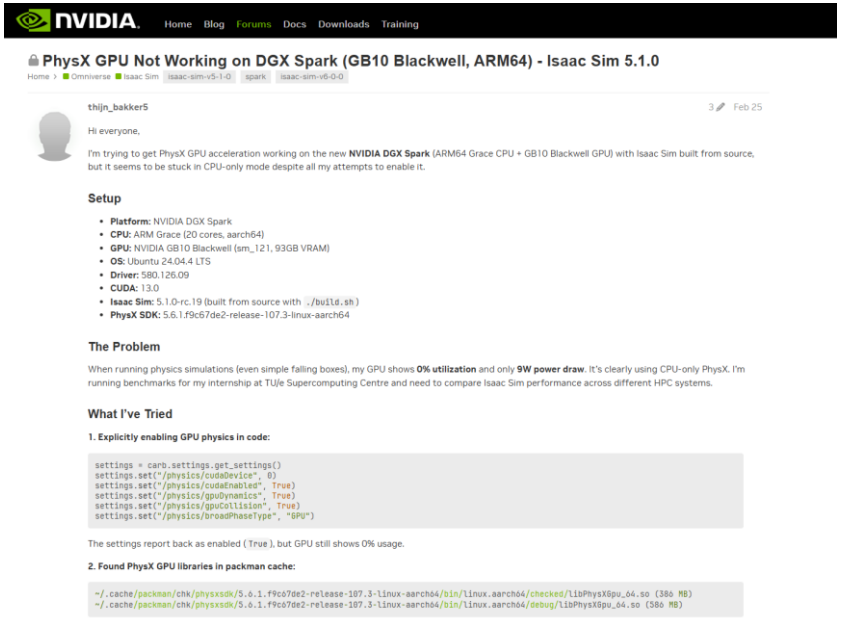
And how I fixed it

Library and Version Issues

PhysX GPU Not Working

0% GPU during physics on all Blackwell GPUs.
Worked with NVIDIA engineer Vlad Khomiakov.
Fix: PhysxSchema configuration (not a driver bug).

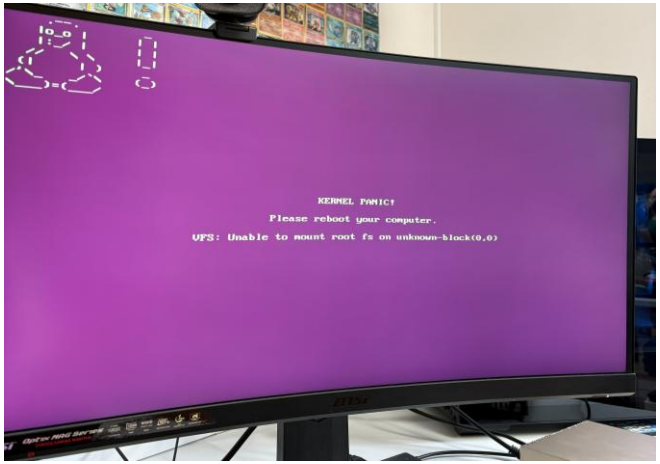


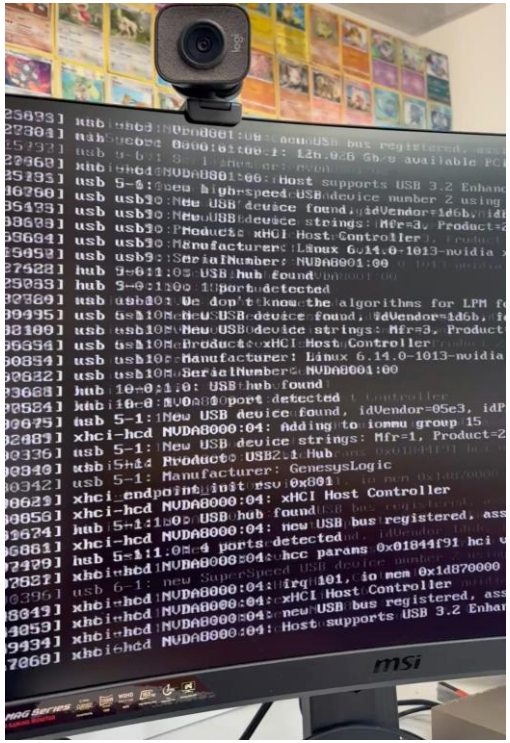


Complete Hardware Failure

DGX Spark Broken

Hardware failure, after running dumb command
Unknown issue
Fix: Full Recovery with USB Flash Drive





DGX Spark System Recovery — Critical Troubleshooting Guide

Severity: CRITICAL — Improper system updates can brick the DGX Spark
Platform: DGX Spark Founders Edition (ARM64 / aarch64)
Official reference: NVIDIA DGX Spark System Recovery
Maintained by: SCC AI Team — TU/e Supercomputing Centre

⚠️ **The #1 Rule: NEVER Run `sudo apt upgrade`**

```
# NEVER RUN THIS ON THE DGX SPARK:
#
#   sudo apt update && sudo apt upgrade
#
# This WILL break the system. It causes a KERNEL PANIC and the
# Spark will no longer boot.
#
```

Running `sudo apt upgrade` on the DGX Spark replaces critical NVIDIA kernel modules and DGX OS components with incompatible versions. The result is a kernel panic on the next boot with the error:

```
KERNEL PANIC!
Please reboot your computer.
VFS: Unable to mount root fs on unknown block(0,0)
```

At this point, the system **cannot boot normally** and requires manual recovery.

05

What I learned

Lessons from the people around me

Key Takeaways from Five Months

01 Test to Find Out!

02 Always Document Everything

03 Talk to Collogues

04 Spread the Word



The Spark has given me way too many headaches...

06

Spreading the word

Just know that it exists!

Conferences & Workshops



NVIDIA from SPARK to SPIKE- Eindhoven



4TU LLM hosting event - Delft

Presentations

BME HUB & PAR

DGX Spark
Isaac Sim
& LLM Hosting

Presented to you by Thijn Bakker

ME Students

ISAAC SIM/LAB
BENCHMARKS

Thijn Bakker
t.t.c.bakker@tue.nl

ME Workshop



Isaac Sim &
Reinforcement Learning

Compute options for robotics research at TU/e

Thijn Bakker

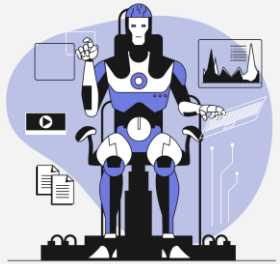
TU/E SUPERCOMPUTING CENTRE – AI TEAM – INTERN

Robotics Lab Workshop · Reinforcement & Imitation Learning · April 28, 2026

ME IT-Committee

Robotics
possibilities
with HPCs

Hardware, software, and where your workloads might fit

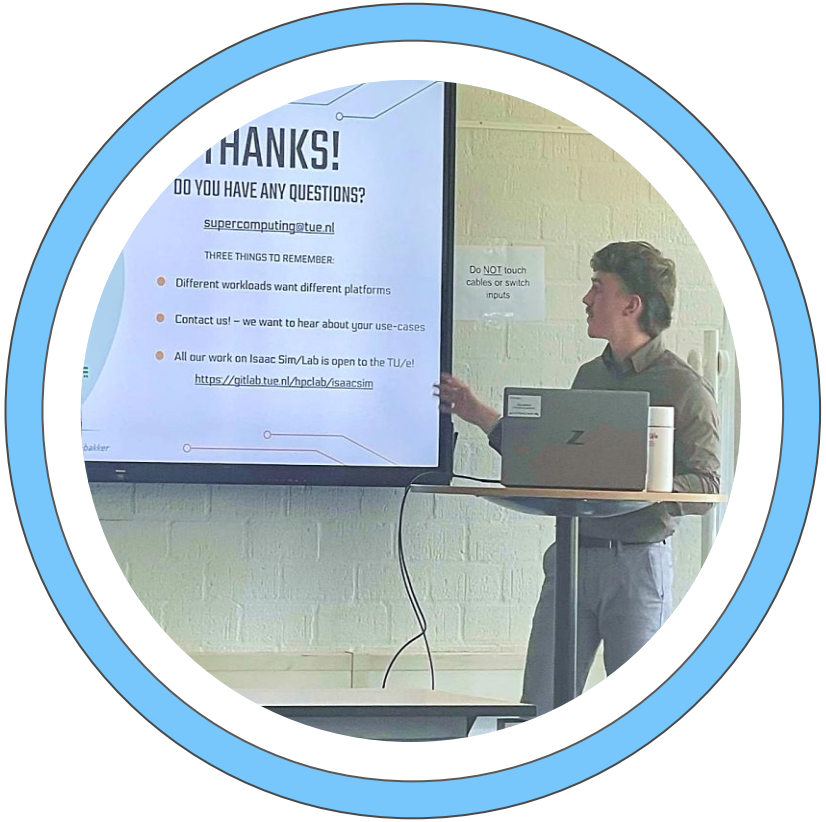


ME IT-Committee Meeting · 21 May 2026

TU/e Supercomputing Centre · Thijn Bakker

"A researcher training a model on their laptop for two weeks doesn't know they could do it in two hours. That's not a technical problem—it's a communication problem."

— Thijn Bakker



07

Impact & What's Next

Results and legacy

From Internship to System Deployment

Benchmarking

✓ Live

Published results
Researchers can choose platforms
Decision-making framework
Spark vs SPIKE-1 vs Cloud vs Snellius

Isaac Lab (Kayden)

✓ Live

Container + Run:AI templates
Kayden trains independently
READMEs in GitLab
Pattern reusable for others

GAIA Support

✓ Live

Research Cloud documented
Ongoing experimentation
Industrial onboarding template
Relationship continues

What Transfers: Documentation, container patterns, processes — infrastructure that outlives the internship. Researchers who require help with robotics on HPCs will be redirected to my documentation!

Thanks!

Nick, Bas, Hengjian, Bram, the whole SCC team, PAR and everyone at TU/e.



I came in as a student with questions. I leave as someone who helped answer questions for others.

4U LLM Hosting Event







PAR Area Party







Umbrella Datacenter







Honorable Mentions

